

Shaiden Trevino Pillay

Junior Full-Stack Developer

Durban, South Africa | +27 62 877 6524 | shaidenpillay15@gmail.com | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Full-Stack Developer who has independently led and contributed to six production systems currently in active use, spanning payroll, biometric attendance, event ticketing, fleet management, and mobile sales analytics, alongside internal platforms for Durban University of Technology and client-facing mobile applications. Comfortable across React 19, Flutter, Node.js, PostgreSQL, Firebase, Prisma ORM, AWS, and WebSockets, with a focus on secure, role-based, real-time business applications with a Bachelor of Information and Communications Technology at Durban University of Technology.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Dart, Python, Java

Frontend / Mobile: React 18/19, Flutter (Riverpod, GoRouter), Android (Java, MVVM), Vite, React Router, Zustand, TanStack React Query, Tailwind CSS, Material UI (MUI), Framer Motion, Recharts / Chart.js

Backend: Node.js, Express.js, Flask (Python), RESTful APIs, WebSockets (Socket.io), JWT & Passport.js authentication, RBAC, IoT device integration

Database & ORM: PostgreSQL, Prisma ORM, Supabase (PostgreSQL), Firebase Firestore, MySQL, SQLite, schema design, query optimization, full-text search

Cloud & DevOps: AWS (S3, Elastic Beanstalk), Firebase Auth, Supabase Auth, Vercel, Render, Docker (basic), CI/CD pipelines, environment configuration

Tools & Practices: Git/GitHub, Postman, VS Code, Jest, Vitest, React Testing Library, Agile development

AI & Integrations: Google Gemini (vision & text), OpenCV, provider-agnostic AI routing (Groq, OpenAI, Ollama), Paystack

PROFESSIONAL EXPERIENCE

Durban University of Technology

Junior Full-Stack Developer - Internal/External Systems Development

Durban, South Africa

June 2025 – Present

Architected and developed enterprise-grade internal business systems and client-facing mobile applications spanning payroll, biometric attendance, pharmacy operations, event ticketing, and mobile sales analytics, using React 19, Flutter, Node.js, PostgreSQL, Firebase, Prisma ORM, WebSockets, and AWS.

PayITSmart - Enterprise Payroll & Employee Management System *Currently in Production*

Lead Full-Stack Developer ·

- Led full-stack development of an enterprise payroll and employee management platform, in production for 21 employees on an hourly/monthly pay cycle, covering onboarding, payroll processing and deductions, and workforce operations.
- Integrated live biometric attendance data from Biometrics24-7 to automatically calculate hours worked and pay for hourly staff, alongside inventory/reservation data and KPI dashboards for operational reporting.
- Designed the PostgreSQL schema and Prisma ORM data layer, and implemented WebSocket-based real-time updates across the platform.
- Reviewed team members' code across the platform, supporting code quality and consistency as lead developer.

Tech stack: React 19, Node.js, PostgreSQL, Prisma ORM, WebSockets

Live: payitsmart.co.za

Biometrics24-7 - Biometric Attendance & Fingerprint Tracking Platform *Currently in Production*

Sole Full-Stack Developer ·

- Built a real-time biometric attendance platform end-to-end, integrating fingerprint scanner hardware via IoT device communication.
- Implemented WebSocket-powered live attendance monitoring, plus analytics and reporting dashboards.
- Developed secure authentication workflows for staff enrollment and attendance verification.

- Architected a dual-mode deployment model: the system runs as a fully standalone app or as a plugin that integrates directly with PayITSmart, feeding live biometric attendance data into payroll calculations to automatically determine employee payslips.
- In production across 7 companies, with each fingerprint scanner supporting up to 127 enrolled fingerprints on-device.

Tech stack: React 19, Node.js, PostgreSQL, WebSockets, IoT Integration

Live: biometrics24-7.co.za

MediIntel - Pharmacy Management System

Full-Stack Developer

- Contributed to frontend and backend development of a pharmacy operations platform covering stock receiving, multi-branch transfers, and returns/expiry tracking.
- Built prescription processing workflows and audit logging for operational reporting and compliance.

Tech stack: React 19, Node.js, PostgreSQL, Prisma ORM

BuzzTicket - Event Management & Ticketing Platform *Currently in Production*

Lead & Sole Full-Stack Developer

- Independently architected and delivered an event booking and ticketing platform, including QR-based ticket generation and validation.
- Integrated Paystack as the payment gateway, creating Paystack subaccounts for event organisers so that ticket revenue is automatically split between the organiser and the system admin at the point of payment.
- Built event analytics dashboards and integrated AWS S3 for secure media and file storage.
- Implemented secure authentication and reporting across the platform.

Tech stack: React 19, Node.js, PostgreSQL, AWS S3, Paystack

Live: buzztickets.co.za

AwehPay - Sales Tracking & Business Analytics Feature

Full-Stack Feature Developer

- Built end-to-end sales tracking and analytics for a Flutter mobile POS app, covering real-time sale recording, daily dashboards, and automated business insights.
- Designed transactional sale-recording logic using Firestore atomic counters (`FieldValue.increment()`) to maintain consistent running totals under concurrent writes.
- Implemented a best-seller/slowest-seller ranking algorithm aggregating quantity sold per product across transaction line items, cross-referenced with live stock data.
- Built a business insights screen with net profit card, custom 7-day income/expense bar chart (no external charting library), product performance cards, and cash flow indicator.
- App published and live on the Google Play Store.

Tech stack: Flutter, Dart, Riverpod, GoRouter, Node.js, Firebase Firestore, Firebase Auth

WhizzFleet - Fleet & Logistics Management Platform

Full-Stack Developer

- In production managing trucks across the Assign and Truck Management modules.
- Built the Assign module's full load-lifecycle tracking (Create → In Progress → POD & Complete) with real-time timestamp capture, an editable-timestamp correction flow, and cross-field ordering validation.
- Designed a custom `DateTimePicker` component and a driver/fleet reassignment workflow that flags conflicts from truck faults or driver leave, with role-based UI gating (Director vs. Controller) enforced server-side.
- Built the Truck Management module, including a server-side registration-expiry alerting system and a fault logging/resolution workflow with severity tracking and fleet-pull integration.
- Implemented a cross-module real-time alert system (sidebar bell badges) kept in sync via React Query cache invalidation across independently owned modules; diagnosed and fixed a cache-key mismatch causing stale counts.

Tech stack: React, TypeScript, Vite, TanStack React Query, Node.js, Express, Prisma ORM, PostgreSQL, AWS S3, Vitest

PERSONAL PROJECTS

NexusOS - Frontend System Architecture

React 19, TypeScript, Zustand, Vite

- Designed a scalable, feature-based frontend architecture for a modular enterprise dashboard, with centralized routing, Zustand state management, lazy loading, and reusable layout systems.

Demo: [View on LinkedIn](#)

SVG Converter - Image to SVG Conversion Platform

React 19, Node.js, Express, Tailwind CSS

- Built a full-stack JPG/PNG-to-SVG conversion tool with drag-and-drop uploads, custom image parsing and SVG generation, and a validated REST API backend.

Facial Recognition Attendance System

Python, Flask, OpenCvD

- Built a role-based facial recognition attendance system with real-time webcam processing, automated attendance tracking, and a Flask backend.

Demo: [View on LinkedIn](#)

Peer-to-Peer Tutoring Platform (Mobile)

Android, Java, MVVM Architecture

- Built a mobile tutoring platform with session scheduling and QR-based attendance tracking, applying MVVM architecture, authentication, and admin workflows.

GovGuide SA - AI-Powered Government Services Platform

React 19, Vite, Node.js, Express, Supabase (PostgreSQL), Google Gemini

- Built a platform that helps South African citizens navigate government processes, starting with NSFAS, consolidating official eligibility rules and document checklists sourced directly from government policy documents, with PostgreSQL full-text search handling most queries without an AI call.
- Built an AI-powered document checker and rejection-letter analyser using Google Gemini vision, reserving AI for the one feature that needs it, and implemented Supabase Auth with row-level security to keep uploaded documents scoped privately to each user.

Demo: [View on LinkedIn](#)

AI Learning Adventure Platform — AI Literacy Learning Game

React, TypeScript, Vite, Tailwind CSS, Zustand, Framer Motion · TopCoder Rapid Application Development Challenge

- Built a 10-mission AI-literacy learning platform (56 quiz questions, 30 flashcards, XP/badge progression) with a full admin CMS for mission, quiz, and flashcard content management.
- Built a provider-agnostic AI tutor supporting Groq, Gemini, Ollama, and OpenAI, with a zero-config mock-mode fallback so the app stays fully functional and free to evaluate.
- Covered core logic with a 76-test suite spanning AI routing, progress/content stores, and component rendering.

Demo: [View demo video](#)

EDUCATION

Bachelor of Information and Communications Technology

2023 – 2026

Durban University of Technology

National Senior Certificate (Bachelor's Pass)

2018 – 2022

Kharwastan Secondary School

CERTIFICATIONS

AWS Certified Developer – Associate (2025) • AWS Lambda Functions, Amazon Web Services (2025)

Get Connected, Cisco Networking Academy (2024) • NDG Linux Unhatched, Cisco Networking Academy (2024)